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NFTransUNet: A Neuro-Fuzzy Deep Learning Model for Breast Tumor Segmentation in Big Data Environment

Ho-Dat Tran[1,2], Thuong-Cang Phan[1], Ngoc-Thuan Bui[3], Vinh-Phong Nguyen[3], and Anh-Cang Phan[2]
[1] Can Tho University • [2] Vinh Long University of Technology Education • [3] Can Tho Oncology Hospital

Abstract

Segmentation of breast tumors from mammograms plays a crucial role in computer-aided diagnosis. However, the diversity in shape, irregular boundaries, and varying sizes of breast tumors remains a significant challenge. This paper proposes **NFTransUNet**, a hybrid deep learning architecture integrating convolutional neural networks, a Transformer-based model, and a neuro-fuzzy inference module. Built upon a fuzzy inference system with Gaussian membership functions, it enables adaptive modeling of complex features while preserving boundary information. Deployed in a distributed Apache Spark environment, NFTransUNet achieved a Dice of **0.935** on the VL-Bre dataset and **0.930** on the VinDr-Mammo dataset, improving performance while maintaining efficient distributed processing.

Key Contributions

Neuro-Fuzzy Integration
Proposal of NFTransUNet, an efficient mammogram segmentation model that integrates a neuro-fuzzy layer with learnable Gaussian membership functions and rule weights into the encoder bottleneck, enabling enhanced feature representation and lesion delineation.

Multi-Backbone Validation
Systematic evaluation of the proposed approach with three backbone architectures (U-Net, U-Net++, and TransUNet) using both the public VinDrMammo dataset and the internal VL-Bre dataset.

Distributed Scalability
An experimental evaluation of the proposed model in an Apache Spark distributed environment, assessing its speedup, parallel efficiency, and scalability through serial-fraction analysis on a multi-node cluster.

Background

Learning-Based Segmentation
Built upon encoder-decoder architectures. U-Net and U-Net++ preserve spatial detail, while TransUNet augments the CNN with Transformers for long-range dependencies.

Neuro-Fuzzy Inference Module
Extends the zero-order Takagi-Sugeno (TSK) fuzzy system. Employs Gaussian membership functions for adaptive refinement:

$$\mu_{j,d}(z_d) = \exp\left(-\frac{(z_d - m_{j,d})^2}{2(s_{j,d} + \delta)^2}\right)$$

Training Objective
Optimized using a hybrid objective consisting of Tversky loss, Binary Cross-Entropy (BCE) loss, and two neuro-fuzzy regularization terms ($\mathcal{R}_{sharp}$ and $\mathcal{R}_{sep}$).

Proposed Method

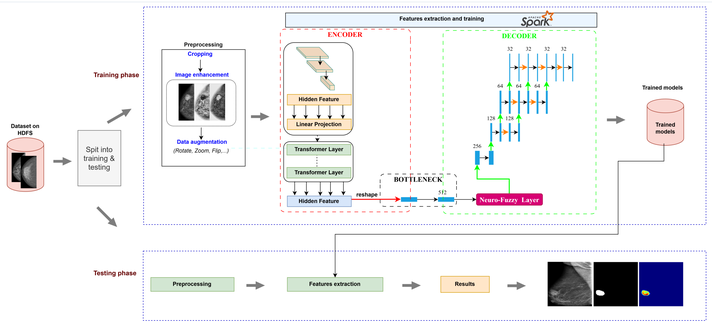


Figure 1: The proposed NFTransUNet pipeline for tumor segmentation on mammograms in a big-data environment.

Experimental Results

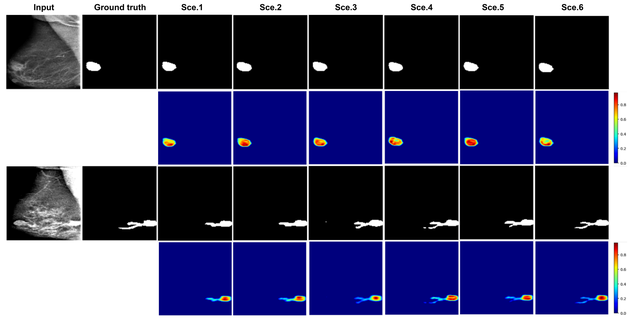


Figure 2: Segmentation and Grad-CAM results across all scenarios.

1

Feature Extraction

Architecture: Couples deep encoder-decoder representations. Data is augmented and passed through a CNN encoder to extract local details, followed by a Transformer block to capture long-range dependencies.

2

Fuzzy Bottleneck

Inference Layer: The deepest features pass through the neuro-fuzzy layer before decoding. It refines the bottleneck through trainable fuzzy rules, expressing graded confidence about ambiguous tumor boundaries.

3

Distributed Training

Apache Spark: Dataset stored in HDFS. End-to-end training is orchestrated across a cluster (1 Master node, 12 Worker nodes). Model parameters are updated locally and synchronized across workers.

Segmentation Performance (Dice Score)

VL-BRE DATASET (IN-HOUSE) **0.935**

VINDR-MAMMO (PUBLIC) **0.930**

Significance: NFTransUNet yields statistically significant gains ($p < 0.01$) over the baseline. Optimal number of fuzzy rules observed at $K = 4$.

Distributed Training Scalability

6.7× to 8.7×

Speedup on 12-node Apache Spark cluster

- ✓ Substantially shortens training time for big data.
- ✓ Parallel efficiency reaches 0.56 - 0.72.
- ✓ Minor accuracy drop (avg 1.8 points) in distributed execution, still maintaining high clinical viability.

Interpretability (Grad-CAM):
The fuzzy-bottleneck variants produce masks whose boundaries more closely follow the irregular contour of the ground truth. NFTransUNet activations concentrate more tightly on the lesion area.

Conclusion

This study integrates a neuro-fuzzy layer at the encoder bottleneck of a deep segmentation network for mammography images. By implementing a fuzzy inference system with learnable Gaussian membership functions, fuzzy-enhanced models improve segmentation efficiency, particularly on ambiguous tumor-containing regions.

The distributed environment significantly reduces training time on large-scale datasets while maintaining highly accurate lesion delineation.

Future Work

- Further testing and validation across multiple external datasets.
- Extending the model to detect and segment other lesion types, such as microcalcifications.

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